Question 3: Simple Savings Interest

Problem Statement

A bank wants to estimate simple interest for a savings account.

Task

Read principal, annual interest rate (percentage), and duration in years.
Calculate simple interest using principal × rate ×

years ÷ 100. Display interest and total amount to two decimal places.

Input Format

• Line 1: principal amount.

• Line 2: annual rate as a percentage.

• Line 3: number of years.

Output Format

• Print the interest and final amount, each to two decimal places.

Constraints

• 0 > principal < 10,000,000

• 0 > rate < 100

• 0 > years < 100

Sample Input

10000

5

2

Sample Output

Interest: 1000.00

Total: 11000.00

1. AIM

To write and execute a COBOL program to calculate **Simple Interest and Total Amount** based on the principal amount, rate of interest, and time, while validating the input values.

2. ALGORITHM

1. Start the program.
2. Declare variables for the principal amount, rate of interest, time, simple interest, and total amount.
3. Get the principal amount from the user.
4. Get the rate of interest from the user.
5. Get the time period from the user.
6. Check whether the principal amount, rate, and time are within the valid ranges.
7. If the inputs are valid:
    - Calculate simple interest using the formula:

    Simple Interest=(P×R×T) /100

    - Calculate the total amount:

    Total Amount=P+Simple Interest

    - Display the simple interest and total amount.
8. Otherwise, display **"INVALID INPUTS"**.
9. Stop the program.

3. PROGRAM

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IDENTIFICATION DIVISION.
PROGRAM-ID. HELLO.

DATA DIVISION.
WORKING-STORAGE SECTION.

01 WS-RT PIC 99.
01 WS-AM PIC 9(8).
01 WS-T PIC 99.
01 WS-AI PIC 9(9)V99.
01 WS-TOTAL PIC 9(9)V99.
01 WS-INTEREST PIC Z(8)9.99.
01 WS-TOTAL_AMT PIC Z(8)9.99.

PROCEDURE DIVISION.

    DISPLAY "ENTER THE AMOUNT"
    ACCEPT WS-AM

    DISPLAY "ENTER THE RATE"
    ACCEPT WS-RT

    DISPLAY "ENTER THE TIME"
    ACCEPT WS-T
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IF (WS-AM > 0 AND WS-AM < 10000000)
   AND (WS-RT > 0 AND WS-RT < 100)
   AND (WS-T > 0 AND WS-T < 100)

    COMPUTE WS-AI =
        (WS-AM * WS-RT * WS-T) / 100

    MOVE WS-AI TO WS-INTEREST
    DISPLAY "INTEREST : "
        FUNCTION TRIM(WS-INTEREST)

    COMPUTE WS-TOTAL = WS-AM + WS-AI

    MOVE WS-TOTAL TO WS-TOTAL_AMT
    DISPLAY "TOTAL AMOUNT : "
        FUNCTION TRIM(WS-TOTAL_AMT)

ELSE
    DISPLAY "INVALID INPUTS"
END-IF

STOP RUN.
```

4. OUTPUT RESULT



Result

Thus, the COBOL program to calculate simple interest and the total amount was written and executed successfully.